THE COLLISION PRIME NUMBER THEOREM

ALEXANDER S. PETTY

Abstract. For a prime p not dividing a positional base b, the collision count $C(b \bmod p)$ records the number of residues r for which the digit function $\delta(r) = \lfloor br/p \rfloor$ satisfies $\delta(r) = \delta(br \bmod p)$. The signed collision count $S(p) = C(b \bmod p) - \lfloor (p-1)/b \rfloor$ is an integer determined by $p \bmod b^2$, taking values in $[-(b-1),\ b-2]$ with the antisymmetry $S(a) + S(b^2-a) = -1$. The $\varphi(b^2)$ values form the *collision periodic table*, a finite invariant of the base.

The centered collision deviation $S^\circ(p)$ is a linear combination of nontrivial odd Dirichlet characters modulo b^2 (primitive odd for prime bases). From this decomposition we prove the *collision prime number theorem* ($\sum S^\circ(p) = O(x \exp(-c\sqrt{\log x}))$, effective for prime bases), the *collision variance* ($\sum |S^\circ|^2 \sim \sigma^2 x/\log x$), and L-value extraction: for prime bases, the periodic table encodes $|L(1,\chi)|$ for all primitive odd characters modulo b^2 (verified computationally for composite bases through $b = 60$).

For base 10, we prove the diagonal–Bernoulli identity $|S_G(\chi)| = |B_{1,\overline{\chi}}|$ (with base 6 by the same method and base 12 verified computationally), giving $|\widehat{c}(\chi)|^2 \propto |L(1,\chi)|^4$: the collision weights are a fourth power of the L-value. The weights live in $\mathbb{Q}(\sqrt{5})$ for base 10, where $\sqrt{5}$ arises from the 5th roots of unity in the character theory of $(\mathbb{Z}/25\mathbb{Z})^*$.

1. Introduction

The digit function $\delta(r) = \lfloor br/p \rfloor$ assigns to each residue $r \in \{1, \ldots, p-1\}$ a digit $d \in \{0, \ldots, b-1\}$. Multiplication by $g \bmod p$ permutes the residues. The collision count $C(g) = \#\{r : \delta(r) = \delta(gr \bmod p)\}$ measures how many residues share a digit with their image under this permutation. It encodes the interaction between the additive structure of the bins and the multiplicative action of g.

The preceding papers [1, 2, 3] established that the collision count decomposes into Dirichlet characters modulo b^2: the collision transform expands $S^\circ(p)$ over odd characters (primitive odd for prime bases),

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and the decomposition theorem [3] factors each coefficient as $\widehat{c}(\chi) = -B_{1,\overline{\chi}} \cdot \overline{S_G(\chi)}/\phi(m)$. (The decomposition is proved for prime bases in [3]; the Fourier analysis extends to all bases, and the identity is verified computationally for composite bases through $b = 60$.)

This paper proves the consequences of that decomposition. The collision PNT (Section 3) is the starting point. The destination is the fourth-moment identity $|\widehat{c}(\chi)|^2 \propto |L(1,\chi)|^4$ (Corollary 5.5), proved through a new algebraic identity between the diagonal character sum and the first Bernoulli number (Theorem 5.3).

2. The Periodic Table

The collision count at the specific multiplier $g = b$ has a closed form. Write $p = b^2 Q_2 + a$ with $0 < a < b^2$ and $\gcd(a, b) = 1$. Define

$$f(a) = \#\{d \in \{0, \ldots, b-1\} : (b+1)da \bmod b^2 \geq b^2 - a\}. \tag{1}$$

Proposition 2.1 (Collision formula)**.** *For a prime $p > b^2$ with $p \nmid b$:*

$$C(b \bmod p) = b \cdot Q_2 + f(a) - 1,$$

where $Q_2 = \lfloor p/b^2 \rfloor$ and $a = p \bmod b^2$. The slice decomposition of [1] *gives b intervals of length p/b^2, each contributing Q_2 or Q_2+1 collisions. The count $f(a)$ records which intervals gain the extra collision; the -1 excludes $r = 0$.*

The signed collision count $S(p) = C(b \bmod p) - \lfloor (p-1)/b \rfloor$ satisfies $S = f(a) - 1 - \lfloor (a-1)/b \rfloor$, depending only on $a = p \bmod b^2$.

Theorem 2.2 (Finite determination)**.** *For any base $b \geq 2$ and prime $p > b^2$ with $p \nmid b$, the signed collision count $S(p)$ is an integer determined by $p \bmod b^2$.*

Proof. $S(p) = f(a) - 1 - \lfloor (a-1)/b \rfloor$ by Proposition 2.1, where $a = p \bmod b^2$. □

This reduces the study of collisions across all primes to a combinatorial problem on a finite group.

Definition 2.3. The *collision periodic table* is the function $\mathcal{T} \colon (\mathbb{Z}/b^2\mathbb{Z})^* \to \mathbb{Z}$ defined by $\mathcal{T}(a) = S(p)$ for any prime $p \equiv a \pmod{b^2}$ with $p > b^2$.

Theorem 2.4 (Antisymmetry)**.** *For every base $b \geq 3$ and every coprime $a \bmod b^2$:*

$$\mathcal{T}(a) + \mathcal{T}(b^2 - a) = -1.$$

Proof. Three identities.

Complementary counting. Let $r_d = (b{+}1)da \bmod b^2$ for $d = 0, \ldots, b{-}1$. For $d \geq 1$: $r_d \neq 0$ (since $\gcd(a(b{+}1), b^2) = 1$). The complement $b^2 - r_d = (b{+}1)d(b^2{-}a) \bmod b^2$. The condition $r_d \geq b^2 - a$ for $f(a)$ and the condition $b^2 - r_d \geq a$ for $f(b^2{-}a)$ are complementary: every $d \geq 1$ satisfies exactly one, except when $r_d = b^2 - a$ (both). Including $d = 0$ ($r_0 = 0$, neither): $f(a) + f(b^2{-}a) = b$.

Floor identity. $(a{-}1) + (b^2{-}a{-}1) = b^2 - 2$, with remainders $R + R' = b - 2 < b$ modulo b, so $\lfloor (a{-}1)/b \rfloor + \lfloor (b^2{-}a{-}1)/b \rfloor = b - 1$.

Assembly. $\mathcal{T}(a) + \mathcal{T}(b^2{-}a) = [f(a) + f(b^2{-}a)] - 2 - [\lfloor (a{-}1)/b \rfloor + \lfloor (b^2{-}a{-}1)/b \rfloor] = b - 2 - (b{-}1) = -1$. □

Corollary 2.5. *$\mathcal{T}$ takes values in $[-(b{-}1),\ b{-}2]$. Exactly $\varphi(b^2)/2$ classes are negative. The mean of $\mathcal{T}$ is $-1/2$.*

Proof. $f(a) \geq 1$ because $d = b{-}1$ always contributes, so $\mathcal{T}(a) \geq -(b{-}1)$. By antisymmetry, $\mathcal{T}(a) \leq b - 2$. Each pair $(a, b^2{-}a)$ sums to $-1 < 0$, so every pair contributes exactly one negative member. □

	0	1	2	3	4	5	6	7	8	9
0	.	0	.	+2	.	.	.	0	.	+8
1	.	−1	.	−1	.	.	.	+1	.	−1
2	.	0	.	−2	.	.	.	+6	.	0
3	.	−1	.	−1	.	.	.	−3	.	−1
4	.	−4	.	0	.	.	.	−2	.	0
5	.	−1	.	+1	.	.	.	−1	.	+3
6	.	0	.	+2	.	.	.	0	.	0
7	.	−1	.	−7	.	.	.	+1	.	−1
8	.	0	.	−2	.	.	.	0	.	0
9	.	−9	.	−1	.	.	.	−3	.	−1

TABLE 1. The collision periodic table in base 10. Row is the tens digit, column is the units digit of $p \bmod 100$. Dots mark non-coprime classes. For any prime $p > 100$, the entry gives $S = C(10 \bmod p) - \lfloor (p{-}1)/10 \rfloor$.

3. The Collision Prime Number Theorem

The centered deviation $S^\circ = S - \overline{S}_R$ removes the class mean (which depends on the last digit $R = (p{-}1) \bmod b$). By Fourier inversion on

$(\mathbb{Z}/b^2\mathbb{Z})^*$:

$$S^\circ(p) = \sum_{\chi \neq \chi_0} \widehat{c}(\chi)\, \chi(p), \tag{2}$$

a sum over nontrivial characters modulo b^2. The antisymmetry $S^\circ(m-a) = -S^\circ(a)$ kills even characters; the centering kills characters whose conductor divides b [2]. For prime bases, the surviving characters are precisely the primitive odd characters [3].

Theorem 3.1 (Collision PNT)**.** *For any base $b \geq 2$:*

$$\sum_{\substack{p \leq x \\ p \nmid b}} S^\circ(p) = O\Big(x \exp\Big(-c\sqrt{\log x}\Big)\Big),$$

where $c > 0$ depends on b (effective for prime $b \geq 3$ by Theorem 3.2; effective for composite bases verified through $b = 60$).

Proof. By (2), the sum equals $\sum_{\chi \neq \chi_0} \widehat{c}(\chi) \sum_{p \leq x} \chi(p)$, a finite sum (at most $\varphi(b^2) - 1$ terms) using only Fourier inversion on $(\mathbb{Z}/b^2\mathbb{Z})^*$, valid at every base. Each character sum satisfies $\sum_{p \leq x} \chi(p) = O(x \exp(-c_1\sqrt{\log x}))$ [5] for any nontrivial χ at fixed modulus b^2. For prime bases $b \geq 3$: the surviving characters are primitive odd (Theorem 3.2 excludes real primitive characters), so c_1 is effective. For composite bases: effectiveness holds provided no real character survives the centering, which is verified for all bases through $b = 60$. □

Theorem 3.2 (Siegel immunity)**.** *For $b \geq 3$, no real primitive character modulo b^2 exists. For prime bases, where the surviving characters are all primitive, c in Theorem 3.1 is effective.*

Proof. A quadratic character modulo p^{2a} (p odd) has conductor p, not p^{2a}: it is the lift of the Legendre symbol. At $p = 2$ with $2a \geq 4$, the conductor is at most 8. Since b^2 is a perfect square and $b \geq 3$, every quadratic character modulo b^2 is imprimitive. □

Theorem 3.3 (Collision variance)**.** $\displaystyle\sum_{\substack{p \leq x \\ p \nmid b}} |S^\circ(p)|^2 = \sigma^2 \cdot \frac{x}{\log x} + O\left(\frac{x}{\log^2 x}\right),$

where $\sigma^2 = \sum_\chi |\widehat{c}(\chi)|^2$.

Proof. $S^\circ(p)$ depends only on $p \bmod b^2$, so the sum decomposes over residue classes. The PNT for arithmetic progressions with fixed modulus gives $\pi(x; b^2, a) = x/(\varphi(b^2)\log x) + O(x/\log^2 x)$. Summing and applying Parseval: $\sigma^2 = (1/\varphi(b^2)) \sum_a |S^\circ(a)|^2 = \sum_\chi |\widehat{c}(\chi)|^2$. □

The collision deviations cancel in the first moment (the PNT) but persist in the second moment (the variance). In base 10: $\sigma^2 \approx 7.26$, giving typical fluctuations $|S^\circ| \approx 2.7$.

4. L-Value Extraction

The periodic table encodes L-function magnitudes.

Theorem 4.1 (L-value extraction)**.** *For prime bases b (and computationally verified for composite bases through $b = 60$), the magnitudes $|L(1,\chi)|$ for all primitive odd characters modulo b^2 are determined by the collision periodic table via:*

1. *Fourier inversion of $\mathcal{T}(a)$ on $(\mathbb{Z}/b^2\mathbb{Z})^*$ recovers $\widehat{c}(\chi)$.*
2. *The diagonal sum $S_G(\chi) = \sum_{n\in G}[\overline{\chi}(n+1) - \overline{\chi}(n)]$ is computed from the diagonal set G [3] (a finite computation depending on the base, not on primes).*
3. *The decomposition theorem gives $|B_{1,\overline{\chi}}| = \phi(m)\,|\widehat{c}(\chi)|/|S_G(\chi)|$, and the classical formula [5] converts to the L-value: $|L(1,\chi)| = (\pi/b)\,|B_{1,\overline{\chi}}|$.*

The system is overdetermined ($\phi(b^2)$ table entries, $\phi(b^2)/2$ character coefficients), providing verification at each step.

This is a new instance of an old pattern. Dirichlet's class number formula extracts $|L(1,\chi)|$ from the arithmetic of ideal classes through a formula involving π. The collision periodic table does the same from the arithmetic of digit collisions. In both cases, a finite object encodes analytic information that is, in principle, determined by the infinite distribution of primes.

k	$\lvert\widehat{c}\rvert$	$\lvert B_{1,\overline{\chi}}\rvert$	$\lvert L(1,\chi_k)\rvert$	$\frac{\lvert\widehat{c}\rvert}{\lvert B_{1,\overline{\chi}}\rvert\cdot\lvert L\rvert}$
2	0.462	4.298	1.350	0.0796
4	0.362	3.804	1.195	0.0796
6	0.238	3.087	0.970	0.0796
8	0.138	2.351	0.739	0.0796
12	0.138	2.351	0.739	0.0796
14	0.238	3.087	0.970	0.0796
16	0.362	3.804	1.195	0.0796
18	0.462	4.298	1.350	0.0796

Table 2. L-value extraction in base 10. The ratio $|\widehat{c}|/(|B_{1,\overline{\chi}}| \cdot |L|) = 1/(4\pi)$ for all 8 primitive odd characters modulo 100.

5. The Diagonal–Bernoulli Identity

The single constant $1/(4\pi)$ in Table 2 has an algebraic explanation.

Lemma 5.1 (Bernoulli–L identity [5])**.** *For a primitive odd character χ modulo m: $|B_{1,\overline{\chi}}| = (b/\pi)\,|L(1,\chi)|$.*

Corollary 5.2 (L-encoding)**.** $|\widehat{c}(\chi)| = (b/\pi\varphi(m))\,|L(1,\chi)| \cdot |S_G(\chi)|$.

The collision weight factors into a multiplicative part ($|L(1,\chi)|$, encoding the class number) and an additive part ($|S_G(\chi)|$, encoding the bin geometry of the digit function). The question is whether these two factors are related.

Theorem 5.3 (Diagonal–Bernoulli identity)**.** *For all 8 primitive odd characters χ modulo 100 (base 10) and all 2 primitive odd characters modulo 36 (base 6):*

$$|S_G(\chi)| = |B_{1,\overline{\chi}}|.$$

Observation 5.4. The same identity holds for all 8 primitive odd characters modulo 144 (base 12), verified computationally to full precision.

Proof. Write $m = 100 = 4 \times 25$ and decompose each primitive odd character as $\chi = \chi_4 \times \psi$ where $\psi = \chi_{25}^k$ with $k \in \{2,4,6,8,12,14,16,18\}$.

Step 1: CRT decomposition of B_1. Using the Chinese Remainder Theorem with $a = 25a_1 + 76a_2 \bmod 100$, $a_1 \in \{1,3\}$, $a_2 \in (\mathbb{Z}/25\mathbb{Z})^\times$: the linear terms vanish by character orthogonality, and $B_{1,\overline{\chi}}$ reduces to:

$$B_{1,\overline{\chi}} = -\sum_{a_1\in\{1,3\}} \sum_{a_2} \left\lfloor \frac{25a_1 + 76a_2}{100} \right\rfloor \chi_4(a_1)\,\overline{\psi}(a_2).$$

Step 2: The collision indicator. Since $\chi_4(1) = 1$ and $\chi_4(3) = -1$, the sum over a_1 reduces to a floor difference: $\lfloor (75 + 76a_2)/100 \rfloor - \lfloor (25 + 76a_2)/100 \rfloor$. This equals 1 when $\{76a_2/100\} \in [1/4, 3/4)$ and 0 otherwise. The set $C = \{a_2 \in (\mathbb{Z}/25\mathbb{Z})^\times : \{76a_2/100\} \in [1/4, 3/4)\}$ has $|C| = 10$.

Step 3: Squared magnitudes. For $r \in (\mathbb{Z}/25\mathbb{Z})^\times$, define the ratio distribution $R_C(r) = \#\{(a,b) \in C \times C : ba^{-1} \equiv r \pmod{25}\}$, and analogously $R_{\mathrm{SG}}(r)$ for the signed diagonal set $\{1^+, 9^+, 3^-, 7^-\}$. Then $|S_G|^2 = 4\sum_r R_{\mathrm{SG}}(r)\,\psi(r)$ and $|B_{1,\overline{\chi}}|^2 = \sum_r R_C(r)\,\psi(r)$.

Step 4: The even projection. Let $f(r) = R_C(r) - 4\,R_{\mathrm{SG}}(r)$. Direct computation of all 20 coprime values of $r \in (\mathbb{Z}/25\mathbb{Z})^\times$ shows that the even part $f_+(r) = (f(r) + f(25-r))/2$ depends only on $r \bmod 5$: $f_+ = 2$ for $r \equiv \pm 1$ and $f_+ = 8$ for $r \equiv \pm 2$.

Step 5: Kernel orthogonality. The kernel of $(\mathbb{Z}/25\mathbb{Z})^\times \to (\mathbb{Z}/5\mathbb{Z})^\times$ is a subgroup of order 5. Any primitive character modulo 25 is nontrivial

on this kernel. Therefore any function depending only on r mod 5 is orthogonal to all primitive characters.

Step 6: Assembly. Since ψ is an even character ($\psi(-1) = 1$), only the even part of f contributes: $\sum_r f(r)\,\psi(r) = \sum_r f_+(r)\,\psi(r) = 0$. Therefore $|B_{1,\overline{\chi}}|^2 = |S_G|^2$ for all 8 characters.

For base 6 ($m = 36 = 4 \times 9$): the CRT decomposition is over $(\mathbb{Z}/4\mathbb{Z})^\times \times (\mathbb{Z}/9\mathbb{Z})^\times$, giving 2 primitive odd characters. The even projection is constant on $(\mathbb{Z}/9\mathbb{Z})^\times$, and kernel orthogonality is immediate. □

Corollary 5.5 (Fourth-moment identity)**.** *For bases $b \in \{6, 10\}$ and all primitive odd characters:*

$$|\widehat{c}(\chi)|^2 = \frac{b^4}{\pi^4\,\phi(m)^2}\,|L(1,\chi)|^4.$$

The collision weights are a fourth power of the L-value.

Proof. By Theorem 5.3: $|\widehat{c}|^2 = (b/\pi\varphi)^2\,|L|^2\,|S_G|^2 = (b/\pi\varphi)^2\,|L|^2\,|B_{1,\overline{\chi}}|^2 = (b^2/\pi^2\varphi(m))^2\,|L|^4$. □

Observation 5.6. The same fourth-moment identity holds at base 12, verified computationally for all 8 primitive odd characters modulo 144.

Remark 5.7. The identity holds for $b \in \{6, 10, 12\}$ and, verified computationally, no other base up to 60. For prime bases $b \le 5$, the ratio $|S_G|/|B_{1,\overline{\chi}}|$ is constant across characters but not equal to 1. For $b \ge 7$, the ratio varies. What distinguishes these three bases is open.

6. The Algebraic Structure of the Weights

The collision weights are not rational numbers.

Theorem 6.1 (Weight decomposition in $\mathbb{Q}(\sqrt{5})$)**.** *For the 4 conjugate pairs of primitive odd characters modulo 100, the squared diagonal sums are:*

$$|S_G(\chi_k)|^2 = A_k + 2\epsilon_k\sqrt{5},$$

where $(A_k, \epsilon_k) \in \{(14, +1),\ (10, +1),\ (14, -1),\ (10, -1)\}$ for $k = 2, 4, 6, 8$. The irrational parts cancel: $\sum_k 2\epsilon_k = 0$.

Proof. The diagonal reduction gives $|S_G(\chi)|^2 = 4\,|\sum_{j\in\{1,3,7,9\}} \overline{\chi}(j)|^2$. For $\chi = \chi_4 \times \chi_{25}^k$, the character values involve 5th roots of unity, and the squared magnitude evaluates through $\cos(2\pi/5) = (\sqrt{5}-1)/4$. The cancellation follows from $\epsilon_k = -\epsilon_{20-k}$. □

The $\sqrt{5}$ arises from the odd prime factor of the base: $10 = 2 \times 5$. For base $b = 2p$ with $p \ge 5$ an odd prime, the collision weights live in $\mathbb{Q}(\sqrt{p})$: base 10 gives $\mathbb{Q}(\sqrt{5})$ (8 characters, 4 complement pairs),

base 14 gives $\mathbb{Q}(\sqrt{7})$ (18 characters). The structure requires at least two complement pairs ($\varphi(b^2)/2 \geq 4$) for the $\sqrt{p}$ component to distinguish them. For base 6 (2 characters, one pair), the weights are rational. The consequences for zero avoidance are developed in [4].

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Email address: `alexander.petty@gmail.com`